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Mid Term - 2

① HP sequences

> HHHPPPH
 - HHPHPHP
 * HPPHHPH
 ✓ HHPHPHP
 ✓ HHHHHHH
 ✓ P P P P P P P

~~Microstates = 272~~

Sequence = HHHPPP HP

Total number of confirmations = 272

Number of macrostates = 2

Macrostates	Degeneracy
-1	8
0	264

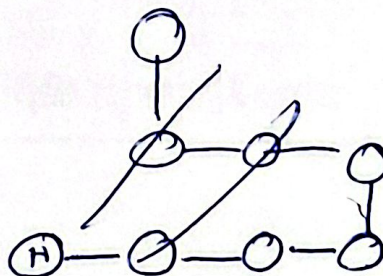
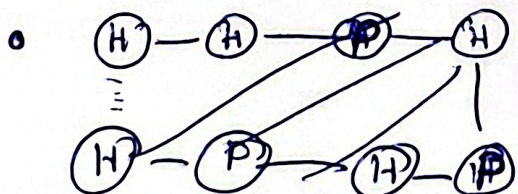
$$\text{Partition function } (Q) = 264 e^{0/k_B T} + 8 e^{1/k_B T}$$

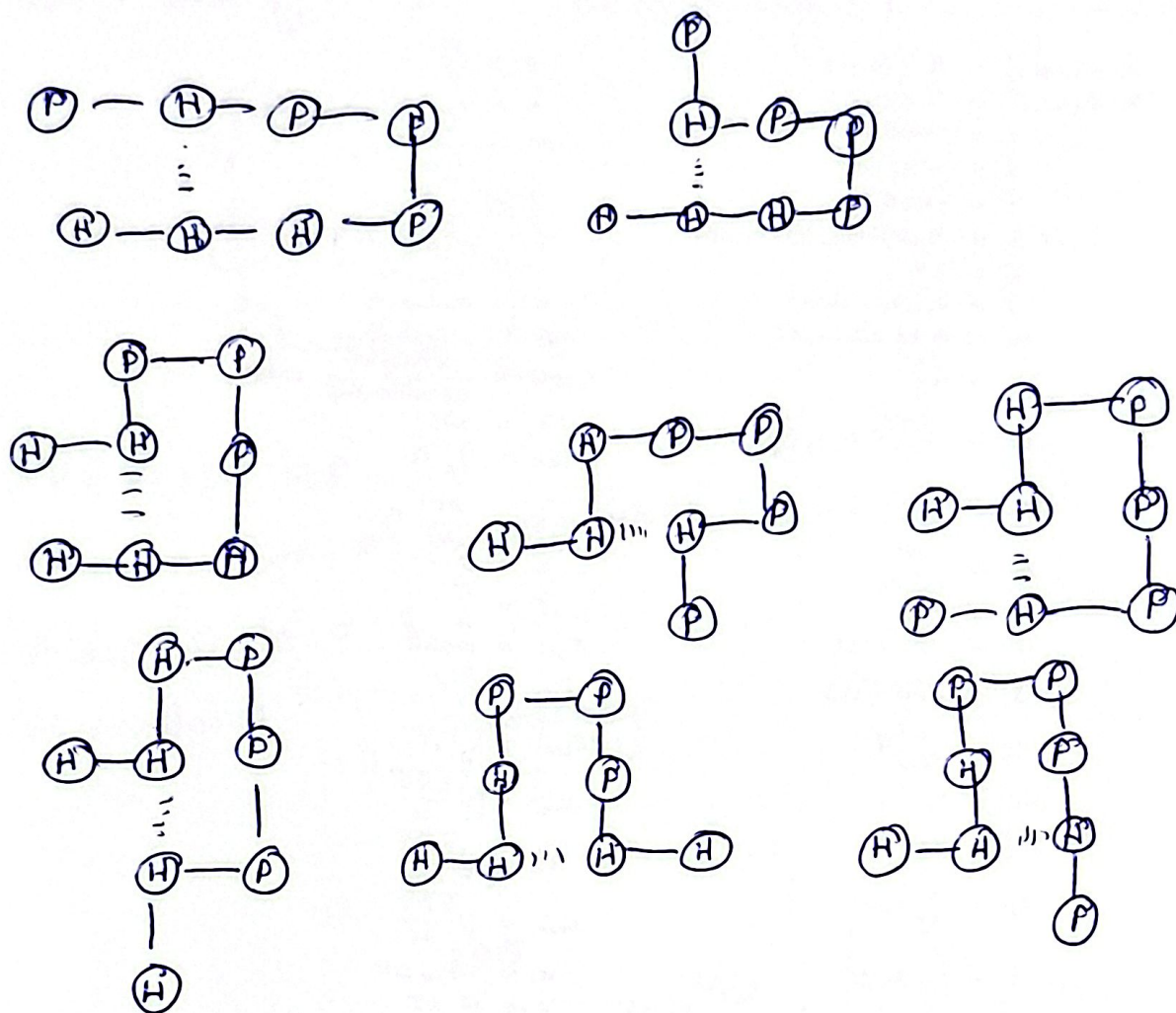
$$= 264 + 8 e^{1/k_B T}$$

$$\text{Average energy} = \frac{-8 e^{1/k_B T} \times k_B T}{264 + 8 e^{1/k_B T}}$$

Lowest energy state energy = -3 Degeneracy = 8

Due to





Sequence = HHPPHPPH

Total number of conformations = 272

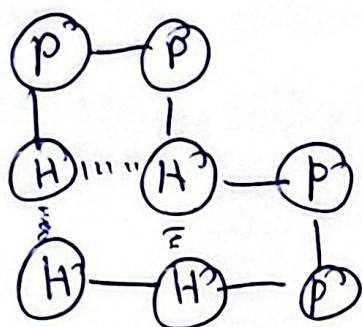
Number of macrostates = 4

Macrostates	Degeneracy
-3	1
-2	7
-1	73
0	191

$$\text{Partition function} = 191 + 73 e^{1/k_B T} + 7 e^{2/k_B T} + 340 e^{3/k_B T}$$

lowest energy = -3

degeneracy = 1



H P P H H P H P

Total number of conformers = 272

Macrostate = 3

Macrostates

Degeneracy

-2

4

-1

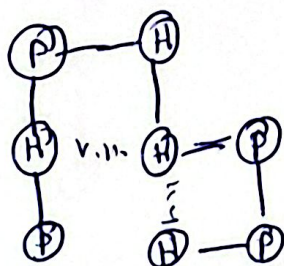
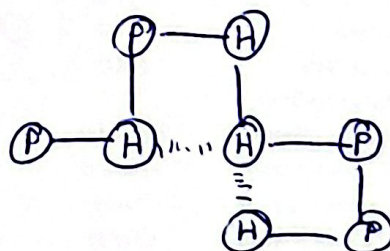
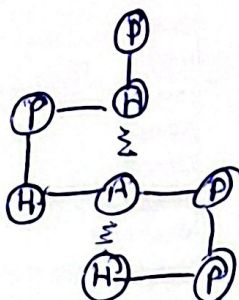
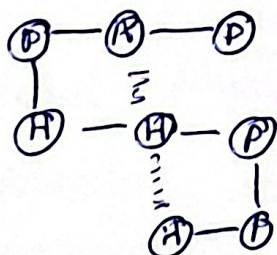
68

0

200

Partition function $Q = 4e^{2/k_B T} + 68e^{1/k_B T} + 200$

Average energy = $\frac{-8e^{2/k_B T} + 68e^{1/k_B T}}{4e^{2/k_B T} + 68e^{1/k_B T} + 200}$



H H P H P H P H

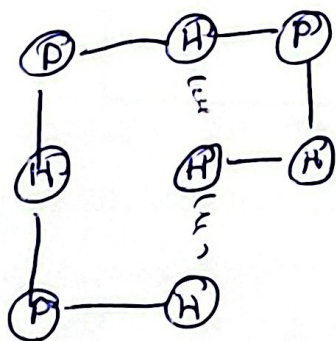
Total number of confirmations = 272

Number of Macrostates = 4

Macrostates	Degeneracy
-3	1
-2	9
-1	54
0	208

$$\text{Partition function } (Q) = 208 + 54 e^{\frac{1}{k_B T}} + 9 e^{\frac{2}{k_B T}} + 1 e^{\frac{3}{k_B T}}$$

$$\text{Average energy} = \frac{-54 e^{\frac{1}{k_B T}} - 18 e^{\frac{2}{k_B T}} - 3 e^{\frac{3}{k_B T}}}{208 + 54 e^{\frac{1}{k_B T}} + 9 e^{\frac{2}{k_B T}} + e^{\frac{3}{k_B T}}}$$



H H H H H H H

Total Number of conformers = 272

Number of Macrostates 4

Macrostates	Degeneracy
-3	19
-2	43
-1	92
0	118

$$\text{Partition function} = 118 + 92 e^{\frac{1}{k_B T}} + 43 e^{\frac{2}{k_B T}} + 19 e^{\frac{3}{k_B T}}$$

$$\text{Average Energy} = \frac{92 e^{\frac{1}{k_B T}} + 86 e^{\frac{2}{k_B T}} + 57 e^{\frac{3}{k_B T}}}{118 + 92 e^{\frac{1}{k_B T}} + 43 e^{\frac{2}{k_B T}} + 19 e^{\frac{3}{k_B T}}}$$

P P P P P P

Total number of conformers = 272

Number of macrostates = 1

Macrostates	Degeneracy
0	272

Partition function = 272

Average energy = 2712

(4) Let's consider for an example ~~HHHHHHHH~~ system. ^{using main_HHH}
adding restrain ^{decreases} partition function ^{decreases} compared to
unrestrained system.
as

Let's consider for an example HHHHHHHH system using
main/HHHHHHHH-restrain-and-unrestrained-comparison.txt file.

In unrestrained system $Z = 4630.30$ while for the restrained
system Z drops to 2890.42 for $k=0.1$, 1864.35 for $k=0.5$,
and 1794.14 for $k=1.0$. This happens because the
restrain adds an extra positive energy for conformations

The restraint also breaks degeneracies. In the Unrestrained
system there are only four energy microstates with
degeneracies -3:19, -2:43, -1:92 and 0:118 after the
restrain is added, these split into new many energy levels
for $k=0.1$ the restrained system has ten distinct energy
values, and similar splitting is seen for $k=0.5$ and $k=1.0$.
this shows the change of degeneracy with restrain.

④ ~~Was sequence - HHHPPPH~~
~~Unrestrained HP energy degeneracy~~
~~Macrostates degeneracy~~
~~-1 8~~
~~0 264~~
 $k = 0.1$
~~Partition function $Z = 115.92002$~~

The restraint also make the polymer more compact
 On average the average radius of gyration decrease
 From 1.2315 in the unrestrained system to 1.1773,
 1.4777 and 1.448 for $k = 0.1, 0.5$, and 1.0 respectively
 The ~~expected~~ Because the ensemble shifts toward
 conformations that both satisfy the HP interactions
 and keep the restrained beads close together.

For the population for unrestrained system,

$$\frac{P_1}{P_2} = e^{-(E_1 - E_2)/k_B T} \quad \text{--- (1)}$$

for the population for restrain system

$$\frac{P'_1}{P'_2} = e^{-(E_1 + E_r) - (E_2 + E_r)]/k_B T} = e^{-(E_1 - E_2)/k_B T} \quad \text{--- (2)}$$

so ~~the~~ now we can see (1) = (2) so that means
 with restrain and unrestrain population ~~ratio~~ ratios
 (relative population) doesn't change.